

Bayesian Analysis Final Project

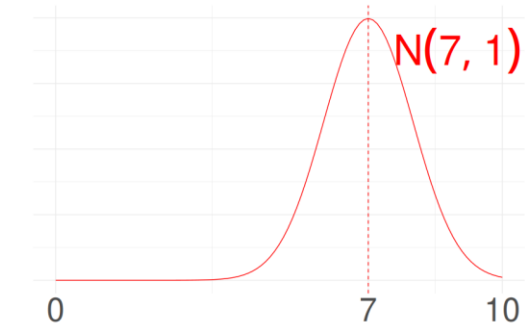
Multi-Armed Bandit Problems, Thompson
Sampling and Route Choice

Introduction: Multi-Armed Bandit Problems

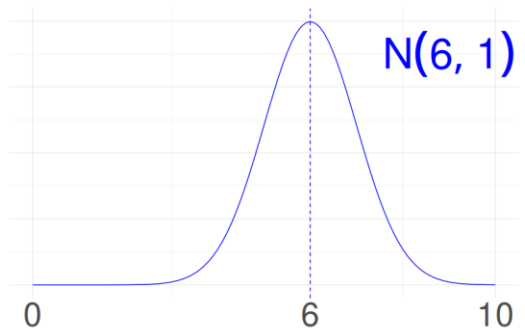
Student



Bayesian



Frequentist



Exploration vs Exploitation

Based on how we do exploration, there are several strategies:

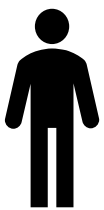
1. Explore only
2. Exploit only
3. Epsilon-greedy
4. **Thompson Sampling**
5. ...

How to evaluate a strategy:

Regret

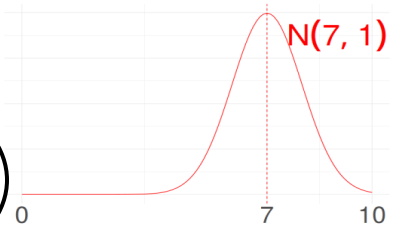
Introduction: Thompson Sampling

Student

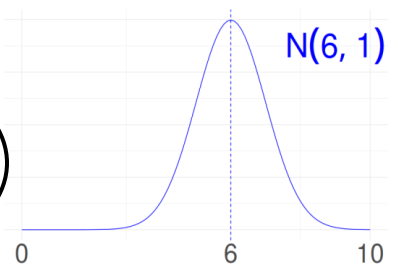


μ unknown
 σ known

1. Bayesian



2. Frequentist



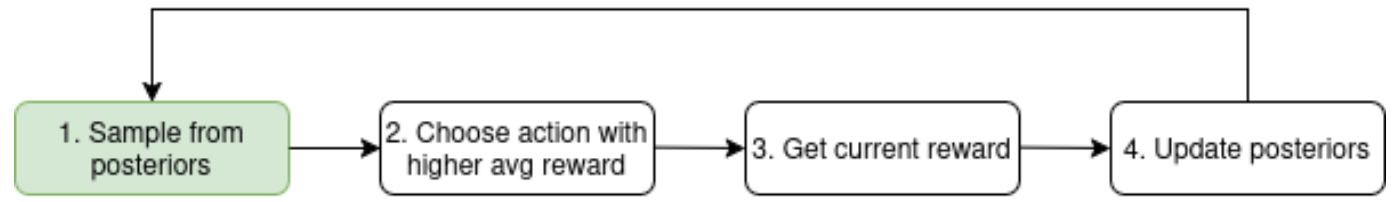
Bayesian model

$$M_i = \{p(s|\mu_i) = N(\mu_i, 1)\}, \pi(\mu_i) = N(0, 100)$$

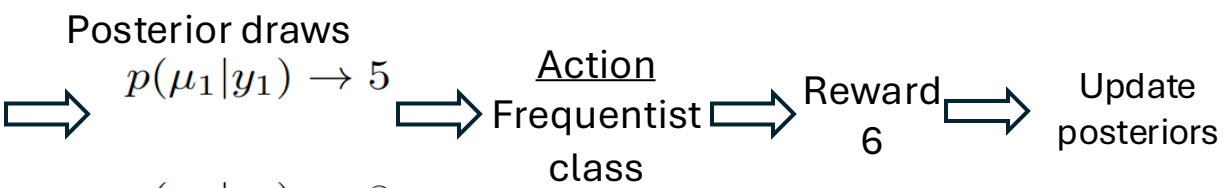
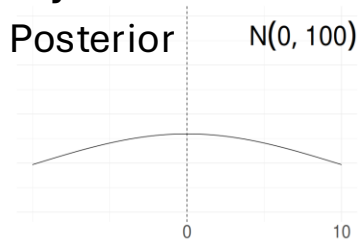
Posterior

$$N(\mu_p, \sigma_p)$$

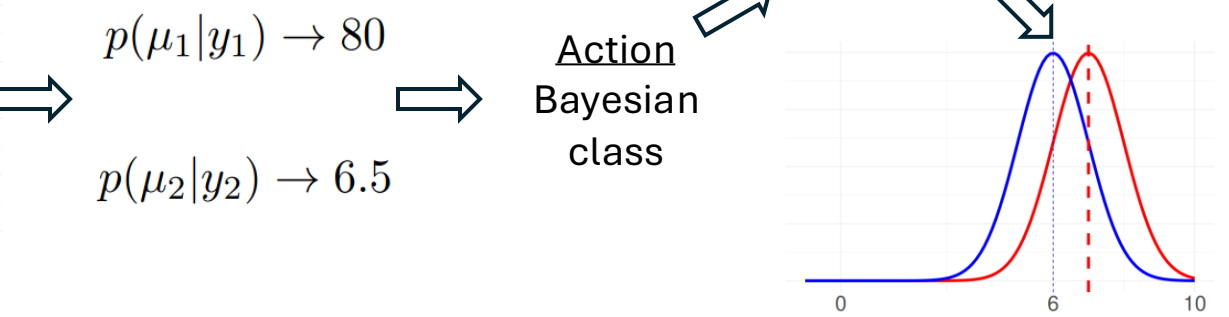
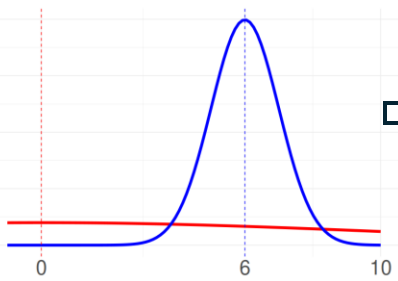
Algorithm



Day 0

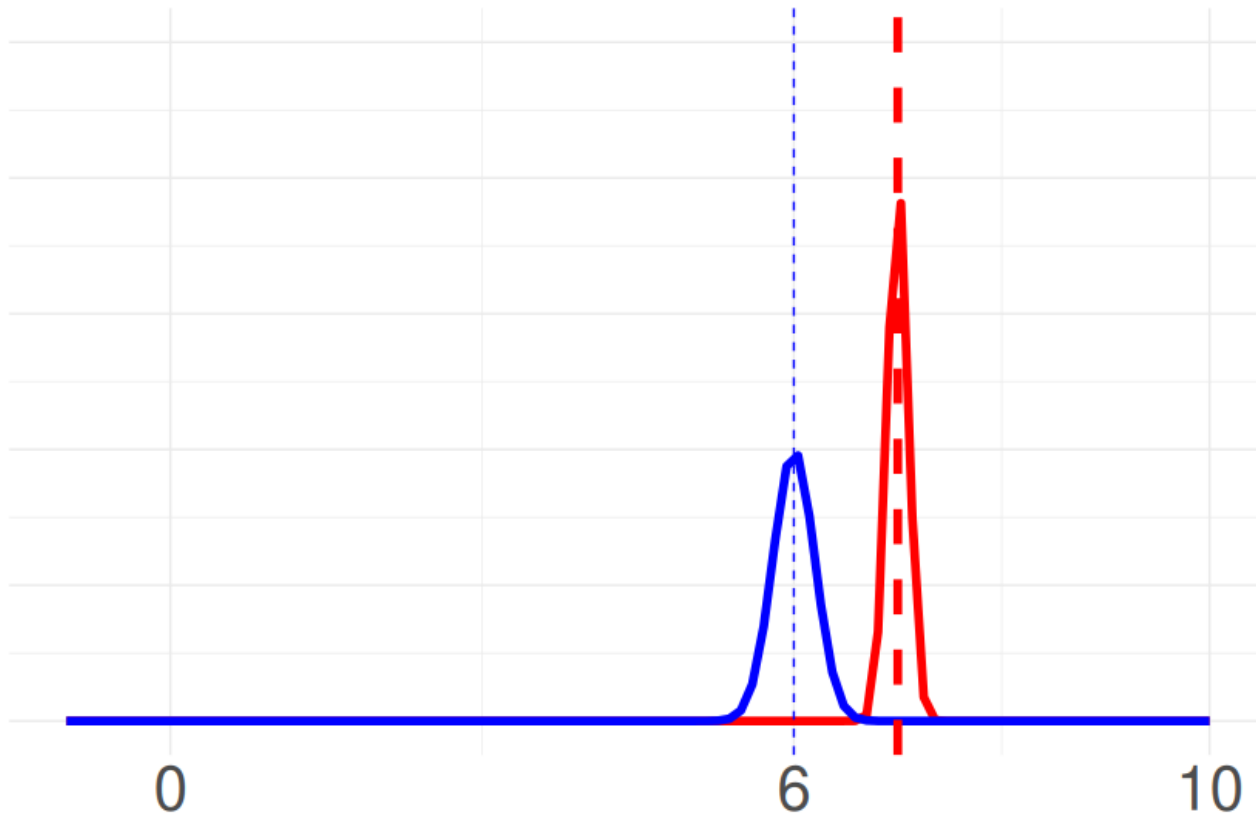


Day 1



Introduction: Thompson Sampling

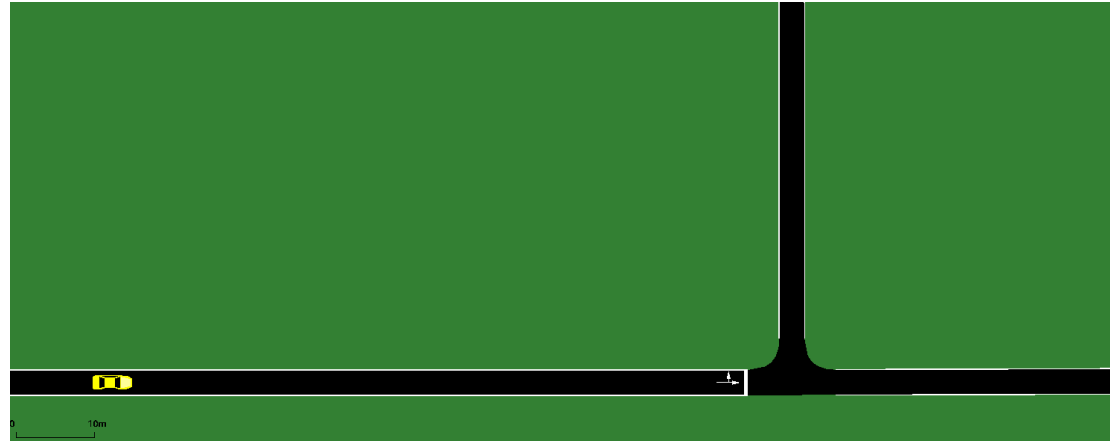
After a few iterations



- Very likely to pick Bayesian class (red) in the next day
- Posterior distribution for Bayesian class mean, will get more and more peaked.
- Hence, finding Bayesian class was the correct answer

Project: Thompson Sampling for Route Choice

Software: SUMO simulator



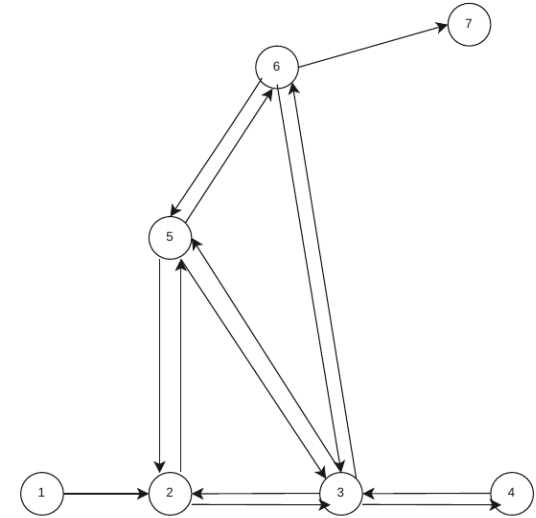
Actions: Routes

Reward: Negative Travel Time

Goal: **Learn which route is the shortest** (has the higher expected reward) **balancing exploration/exploitation** through **Thompson Sampling**.

Data: Travel times of vehicle's agent (generated using the simulator)

Evaluation metric: **Regret** (compares avg reward under optimal strategy with avg reward with our strategy)



Intuition Reward:

Route A = 20 min

Route B = 10 min

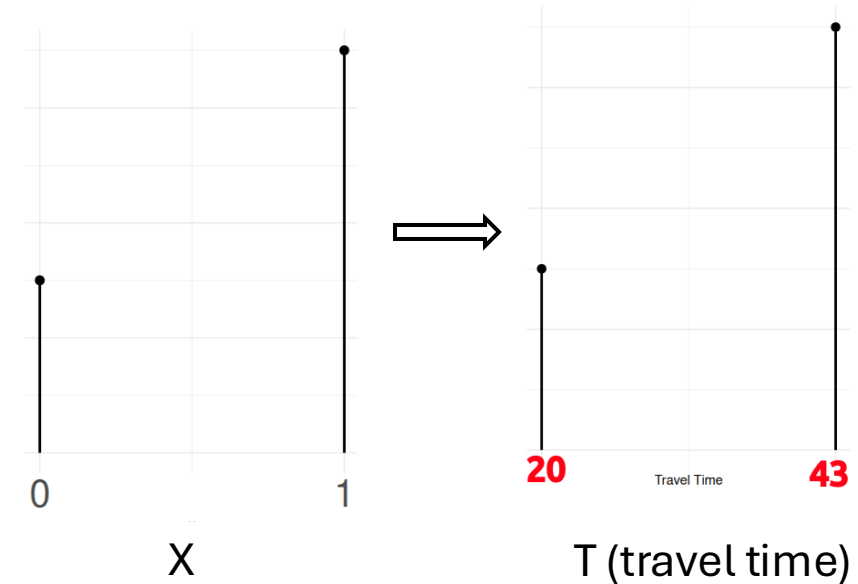
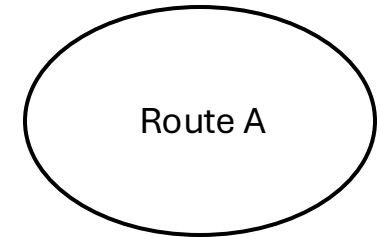
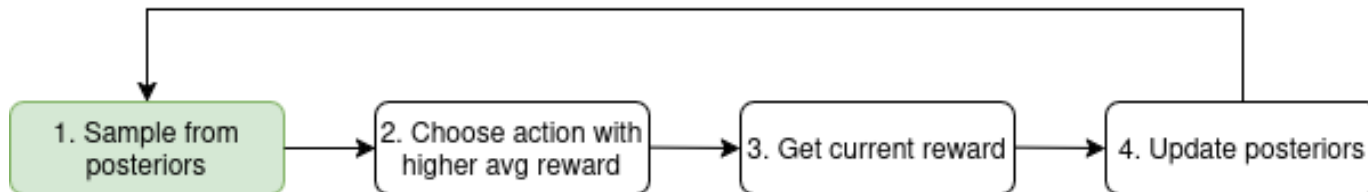
Reward A = -20

Reward B = -10

Project: Thompson Sampling for Route Choice

1st scenario

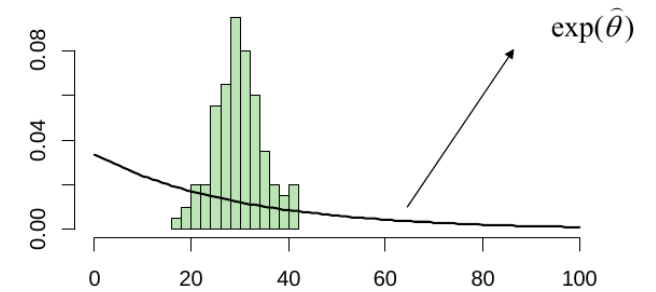
- Hidden reward distributions are **known**
- Only **one vehicle (the agent)**
- **Beta-Bernoulli conjugate model** (each route will have two possible travel times). **Discrete Travel Times.**
- Logic: Very similar to previous example



Project: Thompson Sampling for Route Choice

2nd scenario

- Hidden reward distributions are **unknown**.
- They will be approximated using Monte Carlo Simulation
- Random background traffic plus one agent vehicle
- Workflow:
 1. Environment generation (random traffic). This induces unknown reward distributions for each route.
 2. Run many simulations for each route. The histogram of all the travel times is an **approximation of the unknown reward distribution**.
 3. Model validation (check under which distributions the data is likely to have been generated)
 4. Choose Bayesian Model (appropriate likelihood and non informative **Travel Times**.
Normal-Gamma conjugate prior
 5. Apply same logic than before



Model Validation. Source Bayesian Analysis Slides UPC

$$M_i = \{p(s|\mu_i, \tau_i) = N(\mu_i, \tau_i)\}$$

$$\pi(\tau_i) = \text{Gamma}(\alpha_0, \beta_0)$$

$$\pi(\mu_i|\tau_i) = \text{Normal}(\mu_i, \lambda_0\tau_i)$$